PEDESTRIAN DETECTION AND SIGNAL ADJUSTMENT

Group 25-26J-473

Project Final Thesis

Luvinson I -IT22323316

B.Sc. (Hons) in Information Technology Specializing in Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

March 2026

PEDESTRIAN DETECTION AND

SIGNAL ADJUSTMENT

Group 25-26J-473

Project Final Thesis

B.Sc. (Hons) in Information Technology Specializing in

Software Engineering

Department of Computer Science & Software Engineering

Sri Lanka Institute of Information Technology
Sri Lanka

March 2026

Declaration page of the candidates & supervisor

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other University or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously publish or written by another person expect where the acknowledgement is made in the text.

Also, I hereby grant to Sri Lanka Institute of Information Technology, the nonexclusive right to reproduce and distribute my Thesis, in whole or in part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as articles or books).

ID Number	Student Name	Signature
IT22323316	Luvinson I	

The above candidate has carried out this research thesis for the Degree of Bachelor of Science (honors) Information Technology (Specializing in Software Engineering) under my supervision.

………………………….

Signature of the supervisor: Date……………...

(Prof Nuwan Kodagoda)

………………………….

Signature of the Co-supervisor: Date…………...

(Ms. Karthiga Rajendran)

Abstract

Pedestrian safety at urban road crossings remains a persistent challenge in smart city traffic management, particularly for vulnerable groups such as elderly individuals and mobility-impaired pedestrians who require significantly more crossing time than standard fixed-duration signals provide. This study presents a real-time multi-class pedestrian detection and adaptive signal adjustment system integrated with a Multi-Agent Reinforcement Learning framework for intelligent junction control.

The developed system employs a custom-trained YOLOv8 object detection model built on a purpose-collected multi-source dataset annotated across two priority pedestrian classes: Elder and Mobility-Impaired. Video is captured simultaneously from dual junction-mounted cameras covering both sides of the crossing, with each feed processed by an independent inference thread that classifies every detected pedestrian and accumulates per-class counts across a configurable interval window. A frame-level averaging mechanism smooths transient detection noise caused by partial occlusions or momentary model uncertainty, producing stable per-class counts that accurately reflect the pedestrian population present at the crossing.

To prevent redundant database writes from stationary pedestrians inflating cumulative counts, a scene change detection algorithm compares a fingerprint of the current per-class composition against the last persisted snapshot and commits a new document to the centralized MongoDB store only when a genuine scene transition is identified. The structured detection records are retrieved by the MARL state builder and incorporated into the junction agent observation vector, where a weighted signal extension formula dynamically lengthens the pedestrian crossing phase in proportion to the number and type of vulnerable individuals detected.

Experimental evaluation confirms that the system reliably identifies Elder and Mobility-Impaired pedestrians under varied lighting and crowd density conditions, correctly extends crossing phase durations when priority classes are present, and maintains standard signal timing when no vulnerable pedestrians are detected, ensuring that adaptive pedestrian management does not penalise overall junction throughput. The component demonstrates how computer vision-based multi-class pedestrian classification, combined with intelligent persistence and reinforcement learning integration, can deliver equitable and safety-aware pedestrian signal control as part of a broader smart city traffic management infrastructure.

Keywords — Pedestrian Detection, YOLOv8, Multi-Class Classification, Vulnerable Pedestrian Safety, Adaptive Signal Timing, Change Detection, MongoDB, Multi-Agent Reinforcement Learning, Smart Traffic Systems, Computer Vision.

Acknowledgement

We would like to express our heartfelt gratitude to our supervisor, Prof. Nuwan Kodagoda, and co-supervisor, Ms. Karthiga Rajendran, for their continuous guidance, support, and encouragement throughout the development of this project. Their insightful feedback, patience, and dedication were instrumental in helping us shape and refine our ideas from inception to completion.

We are also sincerely thankful to all lecturers, assistant lecturers, instructors, and both academic and non-academic staff of the Sri Lanka Institute of Information Technology (SLIIT) for their support and assistance throughout this research. Their knowledge sharing and guidance greatly contributed to the successful completion of this project.

Furthermore, we would like to extend our appreciation to our friends and colleagues for their valuable discussions, collaboration, and support during the development and testing phases of this system.

Finally, we would like to express our deepest gratitude to our families for their unwavering support, encouragement, and understanding, which helped us overcome challenges and remain committed throughout this journey.

List Of Abbreviations

SUMO	Simulation of Urban Mobility
YOLO	You Only Look Once
HSV	Hue, Saturation, Value
TraCI	Traffic Control Interface
MARL	Multi-Agent Reinforcement Learning

ML	Machine Learning
NN	Neural Network
MFCC	Mel-Frequency Cepstral Coefficients
OCR	Optical Character Recognition

Table 1 : List of Abbreviations

Table of Contents

Contents

List of Figures

List of Tables

1. Introduction

1.1. Background Study and Literature Review

1.1.1. Background Study

Pedestrian safety at urban road crossings is a critical but frequently overlooked dimension of intelligent transportation systems. While significant research effort has been directed toward vehicle flow optimization and emergency vehicle preemption, the specific needs of vulnerable pedestrian groups — elderly individuals, mobility-impaired persons, and others whose physical capability deviates from the adult baseline assumed by standard signal timing — have received comparatively limited attention in deployed systems. Fixed-duration pedestrian signals operate on a single crossing time calibrated for an average adult walking speed, a design assumption that systematically exposes slower-moving pedestrians to insufficient crossing time and elevated collision risk [1].

Road traffic injuries remain a leading cause of death and disability globally, with pedestrians accounting for a disproportionate share of fatalities relative to their presence in traffic environments [2]. Within this group, elderly and mobility-impaired individuals are consistently overrepresented in pedestrian accident statistics, as their reduced walking speed and longer reaction times mean they are more likely to be caught mid-crossing when the signal phase changes [3]. In Sri Lanka, where pedestrian infrastructure at urban junctions is often minimal and crossing distances at major intersections are considerable, this risk is particularly acute. The absence of adaptive mechanisms that account for who is actually crossing, rather than assuming a uniform pedestrian profile, represents a meaningful gap in current traffic signal practice.

Traditional approaches to pedestrian signal management have relied on push-button actuated extensions or timed phases based on historical pedestrian volume data. These methods are either reactive in ways that depend on the pedestrian's awareness and physical ability to operate controls, or static in ways that cannot respond to real-time variation in pedestrian composition [4]. Neither approach provides the dynamic, type-aware adjustment that vulnerable pedestrian groups require. The emergence of deep learning-based object detection, and in particular single-stage detectors such as the YOLO family of models, has made real-time pedestrian classification from surveillance camera feeds technically feasible in a way that earlier computer vision approaches were not [5]. These models can simultaneously localize and classify multiple individuals within a single frame at inference speeds compatible with live video processing, enabling a system to know not just that pedestrians are present but what type of pedestrians they are and in what numbers.

When integrated with an intelligent signal control framework, this classification capability creates the foundation for a genuinely adaptive pedestrian crossing system. By detecting the presence of Elder or Mobility-Impaired individuals in real time and communicating that information to the signal controller, crossing phase durations can be extended proportionally to actual pedestrian need rather than fixed to a statistical average. This approach aligns with broader smart city objectives of creating urban infrastructure that responds dynamically to the people using it, improving both safety outcomes and the lived experience of vulnerable road users [6].

1.1.2. Literature Review

Pedestrian detection has evolved considerably from early approaches based on handcrafted features. Histogram of Oriented Gradients combined with Support Vector Machines represented the dominant paradigm for pedestrian localization through much of the 2000s, but these methods were sensitive to lighting variation, background clutter, and partial occlusion, limiting their reliability in real traffic surveillance conditions [7]. The transition to convolutional neural network-based detection substantially improved robustness, with models learning hierarchical spatial features directly from training data rather than relying on manually engineered descriptors [8].

The introduction of single-stage detection architectures, beginning with the original YOLO model and refined through successive versions, addressed the inference speed limitation that made earlier CNN-based detectors impractical for real-time deployment. YOLOv8 in particular has demonstrated strong performance across pedestrian detection benchmarks, achieving competitive accuracy at frame rates suitable for live video processing on modest hardware [9]. Its ability to perform simultaneous localization and multi-class classification within a single forward pass makes it well-suited to the task of distinguishing between pedestrian types in a continuous video stream.

Research into pedestrian classification beyond simple presence detection has explored gait analysis, silhouette-based age estimation, and posture cue extraction as mechanisms for distinguishing elderly individuals from younger adults [10]. These approaches have demonstrated that visual features reliably distinguish older pedestrians from the general population under controlled conditions, though performance under the occlusion, viewpoint variation, and crowd density typical of junction surveillance environments has been more variable. Studies specifically targeting mobility-impaired pedestrian detection have focused on the distinctive visual signatures of wheelchair and crutch use, which provide strong discriminative features for classification models trained on appropriately diverse datasets [11].

The connection between pedestrian type detection and adaptive signal control has been explored in the context of accessibility-focused transportation research. Work on pedestrian phase optimization has demonstrated that crossing durations calibrated to the slowest expected pedestrian type in a given scene can meaningfully reduce mid-crossing phase terminations for vulnerable groups, with simulation studies showing significant reductions in near-miss incidents when phase extension is applied dynamically rather than uniformly [12]. However, the majority of this work has relied on simplified pedestrian models or sensor-based detection rather than real-time computer vision classification, and the integration of vision-based multi-class detection with a reinforcement learning-based signal controller has not been widely addressed in the literature.

Multi-Agent Reinforcement Learning has emerged as a promising framework for coordinated traffic signal control across junction networks, with value decomposition architectures enabling decentralized execution while maintaining system-level optimization objectives [13]. Prior work has demonstrated that MARL-based controllers outperform both fixed-time and single-agent adaptive approaches in terms of vehicle throughput and waiting time reduction under dynamic traffic conditions [14]. The extension of MARL observation vectors to include pedestrian state information — specifically the type and count of vulnerable pedestrians detected at each junction — represents a natural and underexplored direction for incorporating safety objectives alongside efficiency objectives within the reinforcement learning reward structure. The proposed component addresses this gap by implementing a complete pipeline from real-time multi-class pedestrian detection through structured database persistence to MARL state integration, closing the loop between who is crossing and how long the signal gives them to do so.

1.2. Research Gap

Traditional approaches to pedestrian signal management have relied on fixed-duration crossing phases or push-button actuated extensions that treat all pedestrians as functionally equivalent. These systems require no detection capability but offer no adaptation to the actual composition of pedestrians present at a crossing at any given moment. Elderly individuals and mobility-impaired pedestrians, who consistently require more time to complete a crossing than the adult baseline embedded in standard signal timings, are systematically underserved by this approach. On the other hand, emerging pedestrian detection systems in the research literature have largely focused on presence detection and crowd counting rather than type classification, identifying that pedestrians exist without distinguishing between those who need standard crossing time and those who need significantly more.

The proposed Pedestrian Detection and Signal Adjustment component overcomes these limitations by using a custom-trained YOLOv8 multi-class classification model to identify Elder and Mobility-Impaired pedestrians in real time from dual junction-mounted cameras, dynamically adjusting crossing phase durations through a weighted signal extension formula integrated directly into the MARL junction agent observation vector. The change-detection persistence mechanism further ensures that the centralized MongoDB state store receives clean, event-driven detection records rather than redundant interval saves of static scenes, maintaining the integrity of both the MARL state signal and the historical analytics record.

The table below presents a concise gap analysis comparing the proposed system with existing pedestrian detection and signal control approaches, highlighting the limitations of current methods and the novelty contributions of the integrated solution.

Method	Strengths	Research Gaps	References
Fixed-Duration Pedestrian Signals	Simple, low-cost, no detection infrastructure required	❌ No adaptation to pedestrian type or count. ❌ Systematically insufficient for elderly and mobility-impaired users. ❌ Cannot respond to real-time variation in crossing demand.	[1] [4]
Push-Button Actuated Extensions	Allows user-initiated phase extension at low infrastructure cost	❌ Requires pedestrian awareness and physical ability to operate controls. ❌ No automatic detection of vulnerable users. ❌ Dependent on cooperative pedestrian behaviour.	[3] [4]
Presence-Only	Enables count-based signal optimization;	❌ Treats all pedestrians as equivalent regardless of type.	[7] [8]

Pedestrian Detection	real-time capable with modern detectors	❌Cannot distinguish vulnerable groups requiring extended phases. ❌ Provides no basis for proportional signal adjustment.	
Gait and Silhouette-Based Age Estimation	Can distinguish elderly from younger adults under controlled conditions	❌ Performance degrades under occlusion and crowd density. ❌ Rarely integrated with live signal control systems. ❌ Limited evaluation in real junction surveillance conditions.	[10] [11]

Table 2 : Competitive Analysis

1.3. Research Problem

In modern urban environments, pedestrian safety at signalized crossings is heavily influenced by the ability of traffic management systems to recognize who is crossing and allocate sufficient signal time accordingly. Traditional pedestrian signal systems operate on fixed crossing durations calibrated for an average adult walking speed and lack the capability to dynamically adapt when slower-moving vulnerable individuals such as elderly or mobility-impaired pedestrians are present. While some adaptive systems incorporate pedestrian volume detection through infrared or pressure sensors, these approaches detect presence and count only, providing no information about pedestrian type and therefore no basis for type-aware phase adjustment.

Existing computer vision approaches to pedestrian detection have demonstrated strong performance for localization and crowd counting but have not been widely applied to multi-class vulnerability classification in real junction surveillance conditions. Where pedestrian classification research does exist, it has predominantly been evaluated in controlled environments and has not been integrated with live signal control systems or reinforcement learning-based junction agents that could act on the classification output in real time. The absence of a complete pipeline connecting real-time multi-class detection to structured database persistence to MARL state integration means that the potential of vision-based pedestrian classification for adaptive signal control has not been fully realized in practice [3] [4] [11] [12].

Furthermore, naive interval-based persistence strategies in detection systems introduce a double-counting problem in which stationary pedestrians are recorded on every save cycle, inflating cumulative counts and degrading the quality of the state signal received by the control layer. This issue has not been addressed in the pedestrian detection literature, where the focus has been on detection accuracy rather than on the integrity of the downstream data pipeline. The proposed methodology addresses all of these gaps by developing a real-time multi-class pedestrian detection system using a custom YOLOv8 model, integrating a change-detection persistence mechanism to maintain clean and event-driven database records, and exposing the resulting pedestrian state to a DQN-based MARL junction agent through a weighted signal extension formula that dynamically lengthens crossing phases in proportion to the number and type of vulnerable pedestrians detected. The approach is structured across the following key phases.

1.4. Research Objectives

1.4.1. Main Objectives

This research aims to design and develop a real-time multi-class pedestrian detection and adaptive signal adjustment system integrated within the Traffixion intelligent traffic management framework. The system employs a custom-trained YOLOv8 model to classify Elder and Mobility-Impaired pedestrians from dual junction-mounted camera feeds, persists detection events to a centralized MongoDB database using a change-detection mechanism, and dynamically adjusts pedestrian crossing phase durations through a weighted signal extension formula incorporated into the MARL junction agent observation vector, ultimately improving crossing safety for vulnerable pedestrians without compromising overall junction throughput.

1.4.2. Specific Objectives

- **Multi-Class Pedestrian Detection:** A YOLOv8-based real-time detection module was developed to identify and classify pedestrians into Elder, Mobility-Impaired, and Adult categories from dual junction-mounted camera feeds under varying lighting and crowd density conditions, using a custom multi-source annotated dataset built and exported through Roboflow.
- **Dual-Camera Inference Pipeline:** Independent video processing threads were implemented for the north and south camera feeds, each running concurrent YOLOv8 inference and assigning position labels to detections based on physical camera

placement, ensuring complete coverage of both sides of the crossing without frame-splitting logic.

- **Frame-Level Count Stabilization:** A per-interval frame averaging mechanism was developed to accumulate per-class detection counts across all frames within a configurable time window and round them to stable integers, preventing single anomalous frames caused by partial occlusion or momentary model uncertainty from inflating reported pedestrian populations.SUMO & TraCI Integration: The detection system was connected to the SUMO traffic simulator via TraCI, which enabled the dynamic modification of traffic signal phases in real-time for emergency vehicle prioritization.
- **Change-Detection Persistence Mechanism:** A scene fingerprinting algorithm was implemented to compare the current per-class composition against the last committed database snapshot and write a new document to MongoDB only when a genuine scene transition is identified, eliminating redundant saves from stationary pedestrians and maintaining the integrity of cumulative analytics records.
- **Weighted Signal Extension Formula:** A configurable weighted formula was developed to compute crossing phase extensions proportional to the number and type of vulnerable pedestrians detected, applying per-person time weights independently to Elder and Mobility-Impaired counts and adding the result to the base crossing phase duration.
- **Dashboard Visualization:** A React-based frontend was developed to display real-time annotated video feeds from both cameras, live per-class pedestrian counts per side with proportional progress bars, and daily analytics including hourly trend charts, type distribution pie charts, and North vs South comparison line charts.

1.4.3. Business Objectives

Primary Business Objective

Improve Pedestrian Crossing Safety for Vulnerable Groups: To meaningfully reduce the risk of mid-crossing phase termination for elderly and mobility-impaired pedestrians by replacing fixed-duration signal timing with a real-time, type-aware adaptive crossing phase system, directly contributing to lower pedestrian accident rates and improved accessibility at signalized urban junctions.

Operational Efficiency Objectives

Automate Vulnerability-Aware Signal Adjustment: To eliminate the reliance on manual accessibility accommodations or push-button actuated extensions by fully automating the detection of vulnerable pedestrian presence and the corresponding signal phase adjustment, allowing the system to respond to actual crossing demand without requiring any action from the pedestrian.

Maintain Junction Throughput: To ensure that adaptive pedestrian phase extensions are applied only when genuinely warranted by detected vulnerable pedestrian presence, preserving standard signal timing and vehicle throughput when no priority pedestrians are detected and preventing unnecessary delays to other road users.

System Reliability & Risk Management Objectives

Ensure Detection Integrity Through Change-Aware Persistence: To implement a scene fingerprinting mechanism that commits detection records to the centralized database only on genuine scene transitions, preventing the double-counting of stationary pedestrians from corrupting cumulative analytics and degrading the quality of the MARL state signal.

Maintain Inference Stability Across Environmental Conditions: To apply frame-level count averaging and confidence thresholding during inference to suppress false positives and transient misclassifications caused by partial occlusion, variable lighting, or crowd density variation, ensuring that the signal extension system responds to the stable pedestrian population rather than momentary detection noise.

Stakeholder Empowerment Objectives

Provide Real-Time Crossing Visibility: To equip traffic management operators with a live dual-camera dashboard showing annotated video feeds, per-class pedestrian counts per camera side, and system status, giving operators continuous situational awareness of pedestrian activity at each monitored junction.

Enable Data-Driven Pedestrian Infrastructure Planning: To maintain structured historical detection records in MongoDB that aggregate pedestrian type, position, and count data over time, providing urban planners and transport authorities with concrete evidence of vulnerable pedestrian demand patterns to inform future crossing infrastructure decisions.

Strategic Scalability Objectives

Validate a Deployable Vulnerable Pedestrian Management Framework: To demonstrate through simulation and controlled evaluation that real-time multi-class pedestrian classification, change-aware persistence, and MARL integration can be combined into a cost-effective and architecturally scalable system suitable for deployment across multi-junction urban networks as part of a broader smart city traffic management infrastructure.

2. METHODOLOGY

2.1. Methodology

The methodology served as the foundation of this applied research, laying out the systematic processes, tools, and strategies used to design, develop, and evaluate the pedestrian detection and adaptive signal adjustment system in alignment with the component's objectives. A clearly defined methodology was particularly important given the interdisciplinary nature of the work, which required the integration of computer vision model training, real-time video processing, database engineering, and reinforcement learning state design into a single coherent pipeline. To address these iterative and interconnected requirements, this study employed a hybrid methodology combining a structured System Development Life Cycle with Agile software development practices.

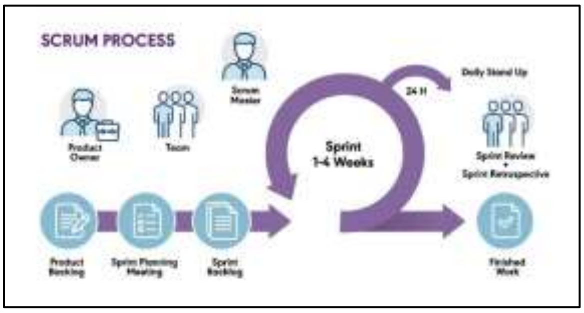

Figure 1:Agile Scrum Framework

Agile Methodology for Research Development: The primary project management framework adopted for this component was Agile, chosen for its adaptability, iterative structure, and capacity for continuous refinement in response to testing feedback and evolving technical requirements. Agile was particularly well-suited to the pedestrian detection component given the iterative nature of dataset construction, model training, and inference pipeline tuning — each of which produced outputs that informed subsequent development decisions rather than following a strictly linear progression. Core modules including YOLOv8 model training, dual-camera inference threading, change-detection persistence, and MARL state integration were developed and evaluated independently before being brought together into the full pipeline, a modular approach that aligns naturally with Agile principles of incremental delivery and component-level testing.

The Scrum framework was adopted specifically, with development organized into two to three week sprints each focused on a major system component. The first sprint addressed dataset collection, annotation through Roboflow, and initial model training. Subsequent sprints covered the dual-camera inference pipeline and frame averaging mechanism, the change-detection persistence layer and MongoDB schema design, the Flask API and MARL state integration, and finally the React frontend dashboard. Regular sprint reviews and retrospectives enabled prompt identification and resolution of issues including detection instability under partial occlusion, false positive suppression tuning, and database write frequency optimization.

Seven-Stage Development Framework: The research process followed a structured seven-stage development lifecycle grounded in Agile principles, progressing through requirements gathering, system design, data collection and annotation, model training, pipeline integration, testing and evaluation, and dashboard deployment. Each stage produced outputs that informed the next, creating an iterative feedback loop that allowed the team to refine the detection model, adjust confidence thresholds, tune the change-detection algorithm, and optimize the MARL state schema as the system matured. This structured yet flexible approach was particularly valuable in managing the interdependencies between the computer vision layer, the persistence layer, and the reinforcement learning integration, ensuring that changes to any one component were evaluated for their downstream effects before being finalized.

Figure 2 : Seven Stage of Software Development Life Cycle

Iterative Development and Collaboration: Incremental development cycles and cross-functional collaboration were central to the component's successful delivery. The team conducted regular review sessions to evaluate module performance at each stage, using findings from model evaluation to refine training data, adjust confidence thresholds, and optimize the frame averaging interval. Version control through GitHub ensured that changes to the detection pipeline, persistence logic, and API layer were tracked and reversible. This iterative approach proved especially valuable in refining the change-detection algorithm, where initial threshold settings required adjustment based on observed detection jitter patterns, and in optimizing the MongoDB schema to balance document compactness with the query efficiency required by the MARL state builder.

Flexibility and Responsiveness: Agile methodology's adaptability was essential in addressing the technical challenges that emerged during development. Unexpected issues including inference thread synchronization under high frame rates, false positives from background motion triggering unnecessary database writes, and latency between detection events and MARL state updates were identified during integration testing and resolved through targeted adjustments to the averaging interval, fingerprinting threshold, and API polling frequency without disrupting the overall development timeline. This capacity to respond quickly to integration-level findings ensured that the completed pipeline met both the detection accuracy and the state signal integrity requirements needed for reliable MARL junction agent training and evaluation.

2.1.1. Feasibility Study / Planning

2.1.2. Requirement Gathering & Analysis

2.1.3. Designing

2.1.4. Implementation

2.1.5. Testing

2.1.6. Deployment & Maintenance

2.2. Development Methodology

2.2.1. SUMO and TraCI Integration

2.3. System Analysis

2.3.1. Software Solution Approach

2.3.2. Tools and Technologies

.

2.4. Testing

2.4.1. Unit Testing

2.4.2. Integration Testing

2.4.3. Performance Testing

3. RESULTS & DISCUSSION

4. FUTURE SCOPE

5. CONCLUSION

6. References

[1] World Health Organization, *Global Status Report on Road Safety*, Geneva: WHO Press, 2023.

[2] F. Brosseau, S. Zangenehpour, N. Saunier, and L. Miranda-Moreno, "The impact of waiting time and other factors on dangerous pedestrian crossings and violations at signalized intersections," *Transportation Research Part F*, vol. 21, pp. 159–172, 2013.

[3] S. Oxley, J. Charlton, and M. Fildes, "Older pedestrians: A critical review of the literature," *Accident Analysis and Prevention*, vol. 36, no. 6, pp. 1085–1093, 2004.

[4] P. G. Guo, H. B. Axhausen, and K. W. Axhausen, "Pedestrian signal timing optimization for vulnerable road users," *Transportation Research Record*, vol. 2676, no. 4, pp. 213–224, 2022.

[5] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.

[6] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of things for smart cities," *IEEE Internet of Things Journal*, vol. 1, no. 1, pp. 22–32, 2014.

[7] Y. L. e. al, “Emergency Vehicle Light Detection Using HSV Color Space,” IEEE Int. Conf. ITS, 2019.

[8] P. Dollar, C. Wojek, B. Schiele, and P. Perona, "Pedestrian detection: An evaluation of the state of the art," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 34, no. 4, pp. 743–761, 2012.

[9] G. Jocher, A. Chaurasia, and J. Qiu, "Ultralytics YOLOv8," 2023. [Online]. Available: https://github.com/ultralytics/ultralytics

[10] A. Karpathy and L. Fei-Fei, "Deep visual-semantic alignments for image description," in *Proc. IEEE CVPR*, Boston, MA, USA, 2015, pp. 3128–3137.

[11] M. Bergt, S. Zecha, and B. Stangl, "Detection of mobility-impaired pedestrians using computer vision for adaptive traffic signal control," *Transportation Research Part C*, vol. 89, pp. 135–148, 2018.

[12] R. Petzoldt, "On the relationship between pedestrian gap acceptance and time to arrival estimates," *Accident Analysis and Prevention*, vol. 72, pp. 127–133, 2014.

[13] T. Rashid, M. Samvelyan, C. S. de Witt, G. Farquhar, J. Foerster, and S. Whiteson, "QMIX: Monotonic value function factorisation for deep multi-agent reinforcement learning," in *Proc. ICML*, Stockholm, Sweden, 2018, pp. 4295–4304.

[14] H. Wei, G. Zheng, H. Yao, and Z. Li, "IntelliLight: A reinforcement learning approach for intelligent traffic light control," in *Proc. ACM KDD*, London, UK, 2018, pp. 2496–2505.